P9 — Error log of PROJECT spacetimes

Provenance: every error made during the work, how it was found, and how it was corrected

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Purpose

Hard Rule 7 of the task specification forbids shortcuts and demands that problems be explained. This file therefore records every error that was made while the project was carried out, including errors that were corrected before anything was published. For each error it states what was wrong, how it was detected, what was done, and where the corrected statement can be checked. An error that is not listed here and is found later is to be added, not silently repaired.

Errors in the construction and verification of the Dirac matrices

No.	What was wrong	How it was found	Correction	Where to check
E1	The first version of check `71-P4` stated the block form of the complex projectors of 7+1 with the wrong sign of the off-diagonal blocks.	The program compares the statement with the computed matrices; the check failed.	The statement was corrected: χ_{1671} has the blocks $(0, -i;\ i, 0)$, and $2P^{L}_{1671}$ has the blocks $(1, i;\ -i, 1)$.	`data/matrix_checks.txt`, line `71-P4`
E2	The 7+1 matrices of the project were first defined as the covariant matrices `Γ17[swap04(A)]` of `7+1.nb`. The Dirac equation of the notebook, however, is written with the contravariant matrices `Γ17up`.	Re-reading sections 5.3 and 7.1 of `7+1.nb` while writing the Dirac equation of dirac1671.	The project's matrices are $\gamma^{A}_{1671} =$ `Γ17up[swap04(A)]`; then ω_{1671} equals `Γ17[8]` exactly and χ_{1671} equals `χ17`.	`data/matrix_checks.txt`, lines `71-N2`, `71-V1`, `71-P1`
E3	Check `71-Q9` first claimed that the two-particle content of $\psi^{T} C^{-}_{1671} \gamma^{j}_{1671} \psi$ is $+\tau^{j}_{1671}$.	The check failed.	The content is $-\tau^{j}_{1671}$.	line `71-Q9`
E4	Check `71-P5` first claimed that the unitary matrix U^{c}_{1671} maps P^{L}_{1671} to P^{L}_{1644}.	The check failed.	It maps P^{L}_{1671} to P^{R}_{1644} and P^{R}_{1671} to P^{L}_{1644}; the reason (two different sign conventions for the chirality matrix in `7+1.nb`) is stated in the check and in Sec. 2.6 of "dirac matrices".	line `71-P5`

No.	What was wrong	How it was found	Correction	Where to check
E5	A commit was created although one check (E4) still failed, because the commands were chained in a way that did not stop on failure; its message claimed that all checks pass.	Reading the output after the commit.	The check was corrected, the program was run again (all checks passed), and the commit was amended before it left the workspace; it was never pushed in the faulty state. Since then a commit is made only if the check program returns success.	`git log`: commit "Dirac equations (dirac1644, dirac1671) and Pauli classification of fermion pairs"
E6	The first implementation of the exact rank (fraction-free elimination with 128-bit integers) overflowed for 256×256 matrices.	The Rust program is compiled with overflow checks and stopped with an error instead of returning a wrong number.	Ranks are computed modulo the prime $2^{61}-1$, with the argument (in the documentation of `rank`) that a rank found modulo a prime is a lower bound of the rank over the rational numbers, so that a full rank is proved; statements about a null space are in addition backed by an explicit null vector. The Python script confirms the same statements by other methods (Gram matrices, enumeration, exact rational elimination).	`code/spacetimes-rs/src/imat.rs`; `data/matrix_checks_independent.txt`
E7	**Symbol collision introduced by this project.** Until November 12, 2026 the program and the drafts used the names `g44`, `g71`, `eta44`, `eta71`, `gamma44`, `gamma71`, `chi71` and the like for the project's matrices. The author's notebooks use these very names for other objects: `g44` and `g71` are metric tensors of the gravity notebooks, `η44` is a 4×4 metric in `Dirac_Nash_Identity.nb`, and `η71`, `Γ71`, `χ71`, `PL71`, `PR71` of `7+1.nb` differ from the project's 7+1 metric, matrices, chirality matrix and projectors. This is exactly the kind of ambiguity that Hard Rule 5 forbids.	A systematic search of all notebooks for names carrying the tags 44, 71 and 17, made while the symbol table of "dirac matrices" was written.	All project symbols were renamed with the tags 1644 and 1671, which occur in no original file (source code, check statements, data files, documents). Other letters with two meanings were separated as well: the 8×8 matrix σ became σ_{88}, the permutation "π" became swap_{04}, the 8×8 blocks "ρ" of 7+1 became τ^{A}_{1671}, the multi-index "K" was replaced by explicit index sets, and rotation parameters are written θ_{AB} instead of ω_{AB}.	MASTER provenance P1; Sec. 0.2 of "dirac matrices"
E8	In the first draft of "dirac matrices" the proof of Lorentz covariance used the transformation $\psi'(x) = S^{-1}\psi(\Lambda x)$ and contracted the indices of Λ incorrectly.	Re-deriving every formula of the draft before the document was assembled.	The correct statement is $\psi'(x) = S\,\psi(\Lambda x)$ with $S\gamma^{A}S^{-1} = \Lambda^{A}{}_{B}\gamma^{B}$; the derivation is given in Sec. 1.8.2. The faulty draft was never committed.	"dirac matrices", Sec. 1.8.2
E9	The first version of the routine of the Python script that counts invariant pair matrices mixed the linear equations of different generators.	Review of the code before its first complete run.	Each generator contributes its own 64 equations for the 28 unknowns. The corrected routine reproduces the numbers of the Rust program (0 and 10 for 7+1; 6, 10, 1 and 0 for 4+4). The faulty version was never committed.	`code/scripts/verify_matrices_independent.py`, function `invariant_dimension`

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E10	The draft of Sec. 1.6 said that cell 166 of `4+4.nb` verifies all five projector relations; the cell verifies three of them.	Comparing the sentence with the cell.	The text now says which three relations the notebook verifies; all five are verified by checks `44-P1`, `44-P2`. Never committed in the faulty form.	“dirac matrices”, Sec. 1.6

Errors in the pairing calculations

No.	What was wrong	How it was found	Correction	Where to check
E11	A formula derived by hand for the exchange amplitude of the radial mode in the two-form pair channels, proportional to $4m^2 - q^2$, was wrong.	The program `spacetimes channels` computes every amplitude numerically from the Dirac matrices; the hand formula disagreed with it.	The amplitude of the radial mode is the same in the vector and in the two-form channels, $(4m^2 + q^2)/4E^2$ on the mass shell; only the phase mode distinguishes the channels. The hand formula was never written into a document.	`data/pair_channel_kernel.csv`; unit test `kernel::tests::hidden_axis_closed_forms`
E12	In the unit test of the angular average of the pairing kernel (`gap.rs`), the reference value was computed with a 64-point Gauss rule, which is inaccurate for the nearly singular integrand; the test failed although the analytic formula was right.	The test failed; refining the reference quadrature made it converge to the analytic value.	The reference is now a composite quadrature graded towards the singularity.	unit test `gap::tests::angular_average_matches_quadrature`
E13	Documentation comments of `gap.rs` and `kernel.rs` contain indented formulas, which the Rust test tool treats as code examples and tries to compile. Commit `97df13f` was made after `cargo test` reported success for the unit tests; the documentation-test stage, which runs afterwards, failed and was overlooked.	Running the complete test command again before the next commit.	The crate declares `doctest = false` (the formulas are not code). All unit tests passed in `97df13f`; no result is affected.	`code/spacetimes-rs/Cargo.toml`
E14	`dimer.rs`: the number of bound states was taken to be the number of nodes of the zero-energy solution inside the integration range; the node that lies beyond the range just above threshold was not counted, which gave the critical coupling $D_c = 1.718$ instead of 1.6798.	The unit test compares D_c with the literature value and failed.	The node of the asymptotic straight line is counted ($u\,u' < 0$ at the end of the range). Never committed in the faulty form.	unit test `dimer::tests::critical_coupling`; `data/dimer.csv`

Errors in the cosmological programs

All errors in this section were found by comparing the Rust solver with the code CLASS (spectra of the author’s NCSS project) *before* any cosmological result was committed; none of them reached a document. They are recorded because the comparison is what makes the final agreement (Supplement S3, Sec. 2) meaningful.

No.	What was wrong	How it was found	Correction	Where to check
E15	`spectra.rs`: wavenumbers above $\max(2.45\,\ell/\tau_0,\ 0.012\ \mathrm{Mpc}^{-1})$ were not allowed to contribute to the multipole ℓ. The projection of a wave of wavenumber k contributes to all $\ell < k\tau_0$ with a slowly decaying weight, so power was lost: the temperature spectrum was too low by up to 14 per cent at $\ell = 100$.	Comparison with CLASS.	The cut was removed; only the cut $\ell > 1.02\,k\tau_0 + 10$ (where the spherical Bessel function is negligible) remains.	`spectra.rs`, comment in `compute`
E16	`bessel.rs`: a unit test demanded an interpolation error of the Bessel table below $3 \times 10^{-7}/x$, which is stricter than the error bound $(h^4/384)\max\lvert f''''\rvert$ of cubic Hermite interpolation with the table's step $h = 0.2$. The code was right and the test was wrong.	The test failed; the measured error was compared with the theoretical bound.	The test now checks the theoretical bounds of value and derivative, and a second test checks the differential equation.	unit tests `bessel::tests::table_interpolation_error_bounds`, `differential_equation`
E17	`boltzmann.rs`: for the matter power spectrum at large wavenumbers the perturbations of the radiation were simply dropped at late times. In the synchronous gauge this is inconsistent: the frame moves with the dark matter, so the radiation has a bulk velocity $-k^2\alpha$ relative to it, and its momentum density is not small compared with that of the baryons. The matter power spectrum was 11 per cent too high for $k > 0.33\ \mathrm{Mpc}^{-1}$.	Comparison of $P(k)$ with CLASS.	Replaced by the radiation-streaming approximation, which keeps these terms. The fast and the full integration now agree to 5×10^{-5} ($k \le 0.42$) and 3×10^{-4} ($k = 2\ \mathrm{Mpc}^{-1}$).	unit test `boltzmann::pk_tests::matter_only_mode_agrees_with_full_integration`
E18	`cosmo.rs`: recombination was computed with the equations of RECFAST 1.0 (factor 1.14 for hydrogen, singlet channel only for helium), which lack the corrections that reproduce the modern multi-level calculations. Effect: 100 θ_s off by 3.4×10^{-4} (relative), spectra off by up to 1.5 per cent at $\ell = 2000$.	Comparison with CLASS (which uses HyRec): a smooth, growing difference in TT and EE after lensing had been accounted for.	The equations of RECFAST 1.5.2 are used: factor 1.125, the two Gaussian corrections of the Lyman-alpha escape factor, and for helium the modified escape probability with hydrogen continuum opacity and the triplet channel. 100 θ_s and r_{drag} now agree with CLASS to 1.3×10^{-5} and 3×10^{-5}.	`cosmo.rs`, `Thermo::new`; unit test `cosmo::tests::background_and_recombination_of_the_fiducial_model`
E19	`cosmo.rs`: while the gas is tightly coupled to the radiation, the temperature equation is extremely stiff. The Runge-Kutta stages advanced the temperature with this stiff derivative; at $z = 2812$, in the first step of the helium rate equation, a stage produced a negative temperature and a NaN, and the program then set the fraction of singly ionized helium to zero. Helium "recombined" within one step instead of gradually between $z = 3000$ and $z = 1700$.	After E18 a difference of 1 per cent at $\ell = 2000$ remained; switching the helium corrections on and off changed nothing at all, which exposed the fault; a printout of $x_e(z)$ confirmed it.	Every stage uses the quasi-static solution of the temperature equation while the coupling is tight (as RECFAST does); the program stops if an ionization fraction is not finite.	unit test `cosmo::tests::ionization_history` (fails for the faulty version)

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E20	`validate_cmb_against_class.py`: the Rust matter power spectrum was interpolated linearly in the logarithms onto the wavenumbers of the reference; near the maximum of $P(k)$, where the Rust grid is coarse, this produced an apparent difference of 1.3 per cent.	The difference was confined to the region where the grid spacing changes.	Cubic-spline interpolation; the difference is 1×10^{-4} there.	`data/pk_validation.csv`
E21	The fiducial amplitude was $A_s = e^{3.0445} \times 10^{-10}$, while the reference spectra of the NCSS project were computed with $A_s = 2.1 \times 10^{-9}$ ($\ln(10^{10} A_s) = 3.04452$): a difference of 2×10^{-5}.	Reading the header of the reference file.	The fiducial model uses $A_s = 2.1 \times 10^{-9}$.	`cosmo.rs`, `Params::fiducial`
E22	A first run of the mock-data fits (`spacetimes forecast`) was started before the validation against CLASS was complete, with the faults E18 and E19 present.	—	The run was stopped and its output discarded; all results in the repository come from the corrected program.	commit `045e161` and later

Errors in drafts of the documents

No.	What was wrong	How it was found	Correction	Where to check
E23	The draft of Supplement S1 stated a closed formula, $F_P = [q^2 - 2(\vec{q} \cdot \hat{n})^2]/4E^2$, for the phase-mode amplitude of a vector pair polarized along an axis $\hat{n}$ of ordinary space, and concluded that its angular average is one third of that of an extra-axis pair.	Before the text was accepted, the formula was compared with `data/pair_channel_kernel.csv`; it is wrong for pairs polarized in ordinary space (their two-particle content depends on the direction of the momentum).	The formula and the conclusion were removed; the text states only what the data show. Never committed in the faulty form.	S1, Sec. 4.2 (iii)
E24	The draft of S1 conjectured that the quartic Ginzburg–Landau invariant is $2(\sum \lvert\Delta^a\rvert^2)^2 - \lvert\sum_a (\Delta^a)^2\rvert^2$ in both spacetimes.	The machine check failed for 4+4 (deviation 4).	The correct invariant contains the metric of the spacetime on the four extra axes, $\lvert\eta_{ab}\Delta^a\Delta^b\rvert^2$; in 4+4 the components along the time-like axes enter with the opposite sign. Never committed in the faulty form.	S1, Eq. (6.2); `data/gl_quartic_check.csv`
E25	The draft of S1 gave the Ginzburg–Landau coefficients a, b and the stiffness K with factors of 2 that did not match the stated normalization of the energy.	Checking that the uniform solution reproduces the textbook result $\lvert\Delta\rvert^2 = \frac{8\pi^2}{7\zeta(3)} T_c^2 (1 - T/T_c)$.	Coefficients corrected: $a = \frac{1}{2} N(0) \ln(T/T_c)$, $b = 7\zeta(3) N(0)/16\pi^2 T_c^2$, $K = 7\zeta(3) N(0) v_F^2/96\pi^2 T_c^2$. Never committed in the faulty form.	S1, Secs. 6.2 and 7